

Clarissa Aurelia Nahid Saputra

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Experience

Triton Robotics

Aug 2025 – Present

Autonomy

San Diego, United States

- Optimized a Kalman Filter pipeline achieving an average converged error of 4×10^{-3} m during jerky, abrupt movements for robust multi-panel center estimation.
- Integrated ROS 2 for perception, ManiSkill for simulation, and FilterPy for probabilistic filtering.

Algoverse

Jul 2025 – Present

Researcher

Remote

- Conducted research on high-norm activations in Diffusion Transformers using PyTorch, analyzing activation distributions across Flux-Schnell, PixArt- σ , and other architectures.
- Performed ablation experiments to evaluate the generative and semantic impact of removing high-norm activations.
- Accepted to XAI4Science at AAAI 2026.

Ruangguru

Jun 2025 – Aug 2025

Software Engineering Intern

Jakarta, Indonesia

- Developed interactive animations for learning Data Structure and Algorithms using React.js and GSAP.
- Built a Node.js CLI tool that syncs animation code with voiceovers using ElevenLabs and Gemini API, automating alignment and boosting efficiency by 90%.
- Integrated Firebase for user authentication and progress tracking.

Projects

memARY: AR glasses that remember everything | Snap AR, ChromaDB, FastAPI, Lens Studio, React.js



- Engineered a real-time camera streaming system that transmits frames from Snap Spectacles to the backend through websockets, enabling continuous perception and semantic processing.
- Integrated Reka's Llama Vision Model for scene summarization and ChromaDB for vector-based memory retrieval, forming a complete cognitive pipeline.

Vanilla Navigation VLA | ROS2, Gazebo, PyDantic, Ollama, Twist Mux



- Built a full autonomous car simulation from scratch in ROS2 and Gazebo.
- Implemented a Vision-Language-Action (VLA) pipeline using a local LLaMA 3.2 Vision model to convert natural language goals into navigation commands.
- Developed modular ROS2 nodes for teleoperation, twist multiplexing, safety stop, and mLLM inference under GPU constraints.

CleanMCP | Claude Hooks, Neo4J, FastMCP, MemMachine.AI



- Implemented intent-driven MCP recommendation, enabling automatic installation and execution of MCP servers based on what the user wants to accomplish.
- Intercepted and logged MCP tool calls into a Neo4j graph database, enabling cross-user, session-level context querying and reuse.

Education

University of California San Diego

2025-2029

Major in Computer Engineering, Minor in Business Economics

San Diego, United States

Awards and Activities

Awards: 7x Hackathon Wins, Honorable Mention IPhO, Honorable Mention APhO

Activities: Influencer with 150k+ followers, Existential Robotics Lab, Data Science Student Society, IEEE

Technical Skills

Languages: Python, C++, C#, Java, Javascript, Typescript

Technologies: Mediapipe, TensorFlow, PyDantic, LangChain, PyTorch, Hugging Face, React.js, Next.js, Figma, Flask, Express.js, Node.js, SQL, YOLO, Scikit-learn, Supabase, Docker, Firebase, MongoDB, OpenCV, Numpy, Pandas, Matplotlib, Puppeteer, Tailwind, BeautifulSoup, RaspberryPi, Arduino, ESP32, ROS